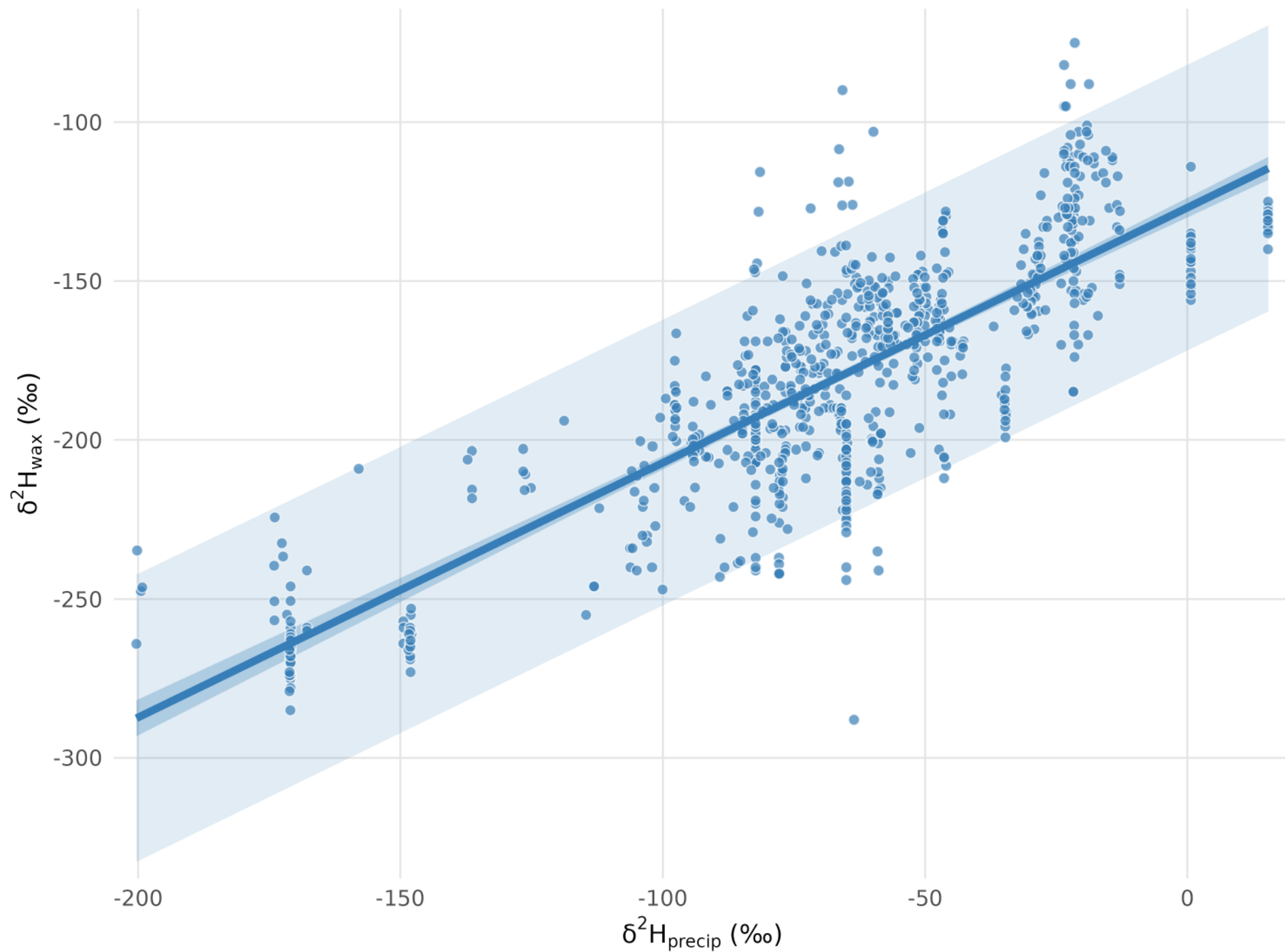


OLS regression (point values)

$y = -126.9 + 0.802x$, $R^2 = 0.653$, $RMSE = 22.8\text{‰}$, $n = 818$



A world map showing annual precipitation in millimeters (mm). The map uses a color scale where lighter shades of yellow and orange represent lower precipitation (100-300 mm), and darker shades of green represent higher precipitation (1000-3000 mm). The map includes latitude markings from 60°N to 60°S and longitude markings from 120°W to 120°E. A legend at the bottom indicates the precipitation levels: 100, 300, 1000, and 3000 mm. The map shows that precipitation is generally higher in tropical regions (near the equator) and lower in arid regions (such as the Sahara and parts of Australia and Asia).

A world map showing the distribution of sampling locations (red dots) across the globe. The map is overlaid with a color-coded elevation scale ranging from 0 meters (green) to 5000 meters (brown). The map includes latitude and longitude markings. The legend indicates the following elevation ranges in meters: 0, 1000, 2000, 3000, 4000, and 5000.

A world map showing the distribution of vegetation types. The map is color-coded: yellow for Grass, light green for Shrub, and dark green for Tree. Red dots are scattered across the map, indicating specific locations. The map includes latitude and longitude markings. A legend in the bottom left corner identifies the colors: Grass (yellow), Shrub (light green), and Tree (dark green).

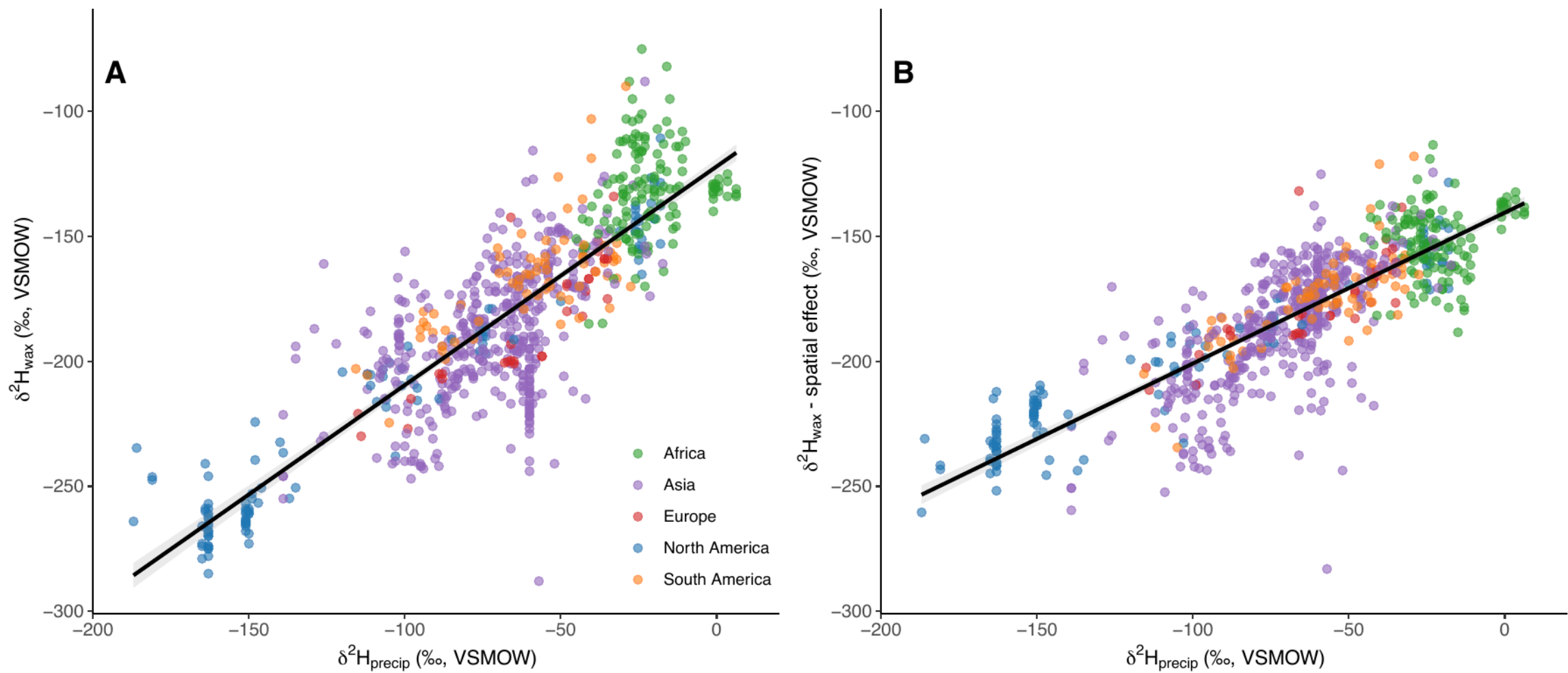
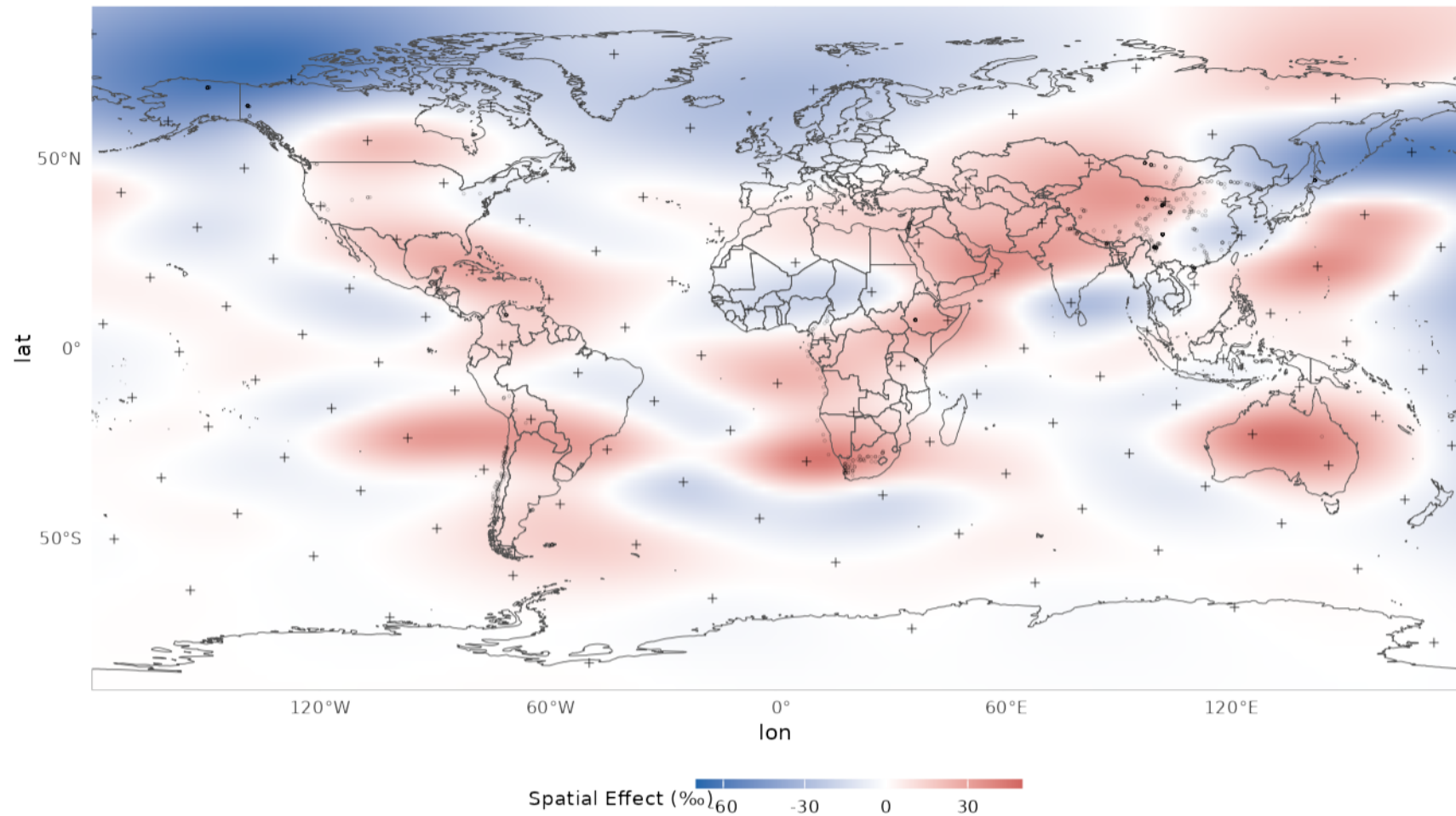


Figure 3: Spatial Patterns in Model Parameters (Full_sp Model)

A. Spatial Intercept Variations

Range: -65.9 to 45.2 ‰ | SD: 16.3 ‰



B. Spatial Slope Variations

Range: 0.440 to 0.458 | Global: 0.448

